

## INTRODUCTION

The digital technology has transformed everyday life, but alongside this has come a new source of Greenhouse Gas (GHG) emissions, the digital Carbon Footprint (CFP).

Common digital activities can account for up to 40% of the per-capita CFP allowance consistent with the 1.5°C climate target.

With approximately 212 million selfies taken every day worldwide, it may be time to zoom in on a single digital action and ask: **What is the environmental cost of a selfie?**

## AIMS / OBJECTIVES

This paper aims to **quantify the carbon footprint (CFP) of a selfie** throughout its entire life cycle, following the methodological principles of ISO 14067:2018.

## FROM SNAP TO CO2

To estimate the carbon footprint of a selfie, seven lifecycle stages and their corresponding energy consumption were identified.

Following ISO 14067:2018, GHG emissions were calculated by multiplying energy consumption by the corresponding emission factors (EFs).

The total CFP, obtained by summing emissions across all stages, was estimated at **0.332 gCO<sub>2</sub>e per selfie** using the global average EF.



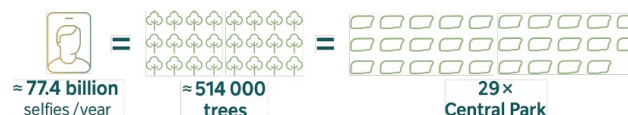
| Process                            | Energy (Wh)  | CFP (gCO <sub>2</sub> e) |
|------------------------------------|--------------|--------------------------|
| Capturing the selfie               | 0.015        | 0.007                    |
| On-device processing               | 0.010        | 0.005                    |
| Uploading (Wi-Fi)                  | 0.135        | 0.064                    |
| Platform processing                | 0.106        | 0.050                    |
| - write to disk                    | 0.012        | 0.006                    |
| - compression                      | 0.014        | 0.007                    |
| - policy check                     | 0.040        | 0.019                    |
| - AI moderation                    | 0.040        | 0.019                    |
| Short-term storage                 | 0.001        | 0.000                    |
| Long-term storage                  | 0.000        | 0.000                    |
| Revisiting / replaying (100 views) | 0.435        | 0.206                    |
| <b>Total</b>                       | <b>0.702</b> | <b>0.332</b>             |

## DISCUSSION

The results confirm that even a single selfie generates measurable greenhouse gas emissions, although uncertainties remain regarding emission factors, processing location, and energy consumption assumptions. The estimated CFP ranges from 0.006 gCO<sub>2</sub>e (Sweden) to 0.497 gCO<sub>2</sub>e (India).

When scaled to the global number of uploaded selfies, the cumulative impact becomes substantial.

**Offsetting the annual selfie footprint** would require approximately 514,000 mature trees, **equivalent to nearly 29 Central Parks**. In arid environments such as Dubai, this would increase to 856,458–1,284,686 trees, corresponding to approximately **15.6–23.4% of the city's total urban tree stock**.



## CONCLUSION

- This study presents **an assessment of the CFP of one selfie** taken and shared on social media.
- Rather than the exact value, the key contribution **is raising awareness that even seemingly trivial digital activities have a measurable environmental impact**.
- The findings highlight the importance of considering digital consumption in both research and climate policy.

## REFERENCES

- Batmunkh, A. Carbon footprint of the most popular social media platforms. Sustainability, 2022.
- Naeem, R. et al. Social media platforms and their digital carbon footprints. Int. J. Media Net., 2023.
- Sharma, P., Dash, B. The digital carbon footprint: Threat to an environmentally sustainable future. 2023.